From Micro to Macro

Definition

LEVELS OF ANALYSIS
- **Micro** : individuals.
- **Meso** : groups, communities, organisations.
- **Macro** : societies, economies, planet.

Macro = used **by many**, **in many places**, **all the time**, **for a long time**.

Key idea

Internet = double-edged sword (Brey 2019) : access to information, communities, commerce $\leftrightarrow$ addiction, overload, harm to democracy. Macro impacts are **controversial** : hard to measure, causality hard to prove.
Problem of one person : involve stakeholders at all levels, work with social sciences, choose your employer.

Example

GenAI and learning (macro) : 26 811 students, 30 months : homework $+18\%$, time -30%, but exams -20% ; worst for junior students and those who outsource.

Explained to a goat

One goat eating a flower is fine ; the whole herd every day for years kills the meadow.

Dis- and Mis-information

Definition
- **Misinformation** : false, **no** intent.
- **Disinformation** : false, **with** intent.
- **Malinformation** : **true**, used to harm.
- **Fake news** : false, in the **style of news**.

Key formulas

Why we believe (Ecker 2022)
Cognitive : System 1 (fast) vs. 2 (slow); intuitive thinking; **illusory truth** (repetition $\Rightarrow$ belief).
Socio-affective : **source cues** (credibility, power, in-group), **false consensus**, emotions, worldview.
Correction : **continued influence effect**.

Example
- A friend : carrots improve night vision $\rightarrow$ **misinformation**.
- Still believing after the debunk $\rightarrow$ **continued influence**.
- Believing Elon Musk's post $\rightarrow$ **source cues**.
- "Fight" boosts reposts 20% $\rightarrow$ **intuitive reactions**.

WEF 2024 : mis/disinformation = most severe global risk for the next 2 years.

Explained to a goat

A goat repeats a false rumour (mis), a fox invents it (dis), or tells a true secret to hurt (mal). Repetition, trusted sources and emotions make it stick.

Software and Disinformation

Key formulas

Creation $\rightarrow$ **Publication** $\rightarrow$ **Propagation**
- **Creation** : content farms, GenAI, deepfakes (volume, speed, realism).
- **Publication** : real users, fake accounts, troll farms, **social bots** (few accounts, much content).
- **Propagation** : resharing (supersharers), click farms, bots, **recommender systems** (engagement, popularity bias).

Key idea

Humans (producers and receivers) + algorithms = socio-technical system with **hidden feedback loops**.

Explained to a goat

A lie is made, posted, then spread by people and by the app's algorithm, in a loop.

Strategy : Edge Cases

Key formulas

Extreme "what if" scenarios once the product is deployed :
- **Global Reach** (**differences**) : languages, cultures, education, regulations, network, time zones.
- **Mass Adoption** (**scale**) : infrastructure, load, security; communities, social norms, democracy.
- **Longevity** (**time**) : prolonged use, feature evolution, societal change, obsolescence.

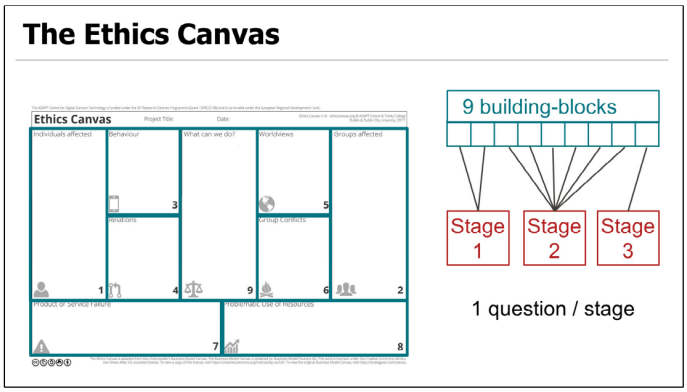
Warning

Graded : extreme scenario + impact on stakeholders + explicit category, then a **mitigation** through design.

Explained to a goat

Everyone everywhere, way too many people, for 20 years : what breaks, who suffers ?

Strategy : Ethics Canvas



Key formulas

9 blocks, 3 stages
Stage 1, who ? 1 individuals (micro), 2 **organised** groups (meso/macro).
Stage 2, impacts ? 3 behaviour, 4 relations, 5 worldviews, 6 group conflicts, 7 product failure, 8 problematic use of resources.
Stage 3 : 9 what can we do ?

Explained to a goat

Nine boxes : who is touched, what changes for them, what we do about it.

Strategy : Causal Loop Diagrams

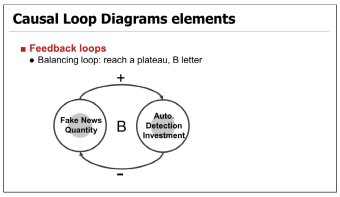
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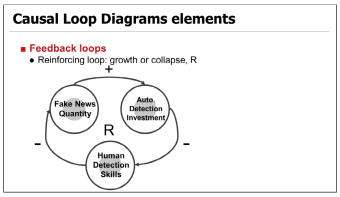
SYSTEM : organised parts interacting as a whole for a purpose. **Systems thinking** : simplify, find patterns, explain behaviour, create solutions.

Key formulas

- **Variables** : quantities, short nouns (no verb).
- $+$ link : same direction ; $-$ link : opposite direction.
- **Delay** : long arrow or **//**.
- **Balancing (B)** : plateau. **Reinforcing (R)** : growth or collapse.

Odd number of $-$ $\Rightarrow$ balancing ; **even** $\Rightarrow$ reinforcing.





Example

Fake news $\xrightarrow{+}$ auto detection $\xrightarrow{-}$ fake news : **B**.
Fake news $\xrightarrow{+}$ auto detection $\xrightarrow{-}$ human skills $\xrightarrow{-}$ fake news : **R**.
Predictive policing : more data in zone A $\rightarrow$ more police $\rightarrow$ more *observed* crime : over-policing in A.

Explained to a goat

Circles and arrows : "+" move together, "−" move opposite. Odd number of $-$: calms down ; even : snowballs.

Weekly Summary

Summary

- Micro / meso / macro ; macro = many, everywhere, always, long.
- Mis (false, no intent), dis (false, intent), mal (true, harm).
- Illusory truth, source cues, false consensus, continued influence.
- Creation $\rightarrow$ publication $\rightarrow$ propagation (bots, recommenders).
- Edge cases : Global Reach, Mass Adoption, Longevity.
- Ethics Canvas : 9 blocks, 3 stages.
- CLD : $+/-$ links, B (odd $-$) vs. R (even $-$).